

HUANXUAN LIAO

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🎓 EDUCATION

Institute of Automation, Chinese Academy of Sciences (CASIA), Beijing 2023 – Present
PhD student in Pattern Recognition and Intelligent Systems (NLPR). Advisors: [Shizhu He](#) and [Kang Liu](#)

North China Electric Power University (NCEPU), Beijing 2019 – 2023
Bachelor of Engineering in Intelligence Science and Technology (Innovation Class). Ranking 1/13

💡 RESEARCH INTERESTS

I work toward **foundation models for long-horizon adaptive intelligence**: models that use growing contexts effectively, retain reusable capabilities, and act reliably over extended interactions.

- **Long-Context Data & Model Co-Design**: long-context data curation and synthesis, length- and capability-aware mixtures, scalable pretraining and mid-training, efficient attention, KV-cache optimization, and architectures to jointly improve long-context capability, scalability, and efficiency ([LCLM-Horizon](#), [HyCo2](#), [Spark](#), [ResAdapt](#)).
- **Knowledge & Continual Adaptation**: acquiring, transferring, internalizing, and updating knowledge and capabilities across evolving tasks, while balancing plasticity, retention, and generalization ([TAGI](#), [AAG](#), [NesyCD](#), [SKIntern](#), [SpaRTA](#)).
- **Agent Learning & Self-Improvement**: persistent memory, long-horizon decision-making, and learning from interaction and feedback to verifiably refine agent memory, policies, and learning procedures ([CITI](#), [KATE](#), [OPERATE](#), [Distill Yourself](#), [SLE](#)).

👥 INTERNSHIP/PROJECT EXPERIENCE

Tencent (Project UP) Beijing Jul. 2026 – Present
Pretraining Long-Context Data and Architecture

- **Long-context pretraining data**: Designed MRCR-style multi-QA samples with denser answer supervision and a final long-range query. Delivered **186.8B tokens** across six 8K–512K bins and a separate **10B-token** training set for controlled comparison; validated answer positions, loss masks, and length distributions.
- **Data recipe and training analysis**: Compared ten recipes under matched training settings across 8K–128K. In a 10B-model ablation, per-answer endings and a final QA improved peak aggregate MRCR from **0.854 to 0.911**; all-multi-QA data degraded at longer lengths, informing mixture design.
- **Sparse architecture exploration**: Compared long-context quality and prefill/decode cost under matched models and budgets; found that lower selector FLOPs alone did not reduce end-to-end cost because of fixed selection overhead and KV movement.
- **Vision-code pretraining data**: Contributed to SVG, sketch-to-HTML, and CAD datasets through image-code checks, cross-library perceptual deduplication, and layered Qwen3-VL-Embedding / SVGFaith-Score screening. The SVG corpus contained **8.13M samples (about 41B tokens)**.

ByteDance Beijing Jun. 2025 – Jun. 2026
Efficient Multimodal Modeling Visual Budgeting and Compression

- **Production AIGC detection**: Deployed efficient multimodal representations for TikTok AIGC detection, increasing online F1 by > 3% while cutting compute by **30%–50%**; reused the representations for embedding retrieval.
- **Resolution before encoding**: Proposed [ResAdapt](#) and [CAPO](#) to allocate a visual budget per frame before encoding. Removed > 90% of visual tokens, processed up to 16× more frames at matched compute, and improved long-video reasoning by > 15% relatively.
- **Compression and architecture evaluation**: Reproduced and compared visual-token and KV compression methods, measuring accuracy against compute; explored skip-layer and dynamic-boundary designs.
- **Long-horizon agent evaluation**: Built [OPERATE](#) to evaluate when an agent should observe, intervene, or hold a plan in evolving simulators, with evidence-based scoring and same-seed counterfactual checks.

- **Hybrid context compression:** Proposed **HyCo₂** to preserve global semantics and local evidence; achieved a 13.1% average gain across seven knowledge-intensive QA tasks while matching uncompressed performance with about 88.8% fewer input tokens.
- **Long-context research:** Core contributor to the [LCLM-Horizon survey](#) on data, architectures, training, and evaluation; maintained [Awesome-LLM-Long-Context-Modeling](#).

📖 SELECTED PUBLICATIONS [FULL LIST]

First-authored papers are listed first. * denotes equal contribution.

- [Huanxuan Liao](#), Yao Xu, Shizhu He, Jun Zhao, Kang Liu. [HyCo₂: Hybrid Context Compression with Global Semantics and Local Evidence](#). (TOIS 2026, accepted)
- [Huanxuan Liao](#), Zhongtao Jiang, Yupu Hao, Yuqiao Tan, Shizhu He, Ben Wang, Jun Zhao, Kun Xu, Kang Liu. [ResAdapt: Adaptive Resolution for Efficient Multimodal Reasoning](#). (arXiv 2026)
- [Huanxuan Liao](#), Shizhu He, Yupu Hao, Yequan Wang, Wenhao Teng, Xiangwen Liao, Jun Zhao, Kang Liu. [Spectral Disentanglement: Rank-Aware Task Adaptation for Rehearsal-free Continual Learning in LLMs](#). (ACL 2026 Main, CCF-A)
- [Huanxuan Liao](#), Yixing Xu, Shizhu He, Guanchen Li, Xuanwu Yin, Dong Li, Emad Barsoum, Jun Zhao, Kang Liu. [Spark: Query-Aware Unstructured Sparsity with Recoverable KV Cache Channel Pruning](#). (AAAI 2026, CCF-A)
- [Huanxuan Liao](#), Shizhu He, Yao Xu, Yuanzhe Zhang, Kang Liu, Jun Zhao. [Neural-Symbolic Collaborative Distillation: Advancing Small Language Models for Complex Reasoning Tasks](#). (AAAI 2025, CCF-A)
- [Huanxuan Liao](#), Shizhu He, Yupu Hao, Xiang Li, Yuanzhe Zhang, Jun Zhao, Kang Liu. [SKIntern: Internalizing Symbolic Knowledge for Distilling Better CoT Capabilities into Small Language Models](#). (COLING 2025, CCF-B)
- [Huanxuan Liao](#), Shizhu He, Yao Xu, Yuanzhe Zhang, Kang Liu, Shengping Liu, Jun Zhao. [Awakening Augmented Generation: Learning to Awaken Internal Knowledge of Large Language Models for Question Answering](#). (COLING 2025, CCF-B)
- [Huanxuan Liao](#), Shizhu He, Yao Xu, Yuanzhe Zhang, Yanchao Hao, Shengping Liu, Kang Liu, Jun Zhao. [From Instance Training to Instruction Learning: Task Adapters Generation from Instructions](#). (NeurIPS 2024, CCF-A)
- Jiaheng Liu, Dawei Zhu, Zhiqi Bai, Yancheng He, [Huanxuan Liao](#), et al. [A Comprehensive Survey on Long Context Language Modeling](#). (arXiv 2025, **Core Contributor**)

♥ AWARDS AND SCHOLARSHIPS

National Scholarship ×3

2021 / 2022 / 2025

Beijing Outstanding Graduate

2023.06

💡 PROFESSIONAL SERVICES

Conference Reviewing: ACL ARR 2024–2026. NeurIPS 2025–2026. ICML 2026. COLM 2026. ICLR 2025–2026. AAAI 2026.

Community: [Awesome-LCLM](#) (Curator & Maintainer; GitHub Stars > 2.2K); LCLM-Horizon survey (core contributor); [NVIDIA/kvpress](#) (contributor); [Distill Yourself](#) (agent memory & preference distillation).